

Running Local vs Running on Production

Backend connection notes for the Invent React + Django inventory app.

What this document is about

The frontend was loading, and the backend server was running, but the app still looked disconnected. The reason was not one single frontend problem. The local Django API first rejected local hosts, then the /assets endpoint crashed because the database was missing a pending migration.

What I Changed

- Updated backend/.env so local development hosts are allowed together with the production DigitalOcean host.
- Confirmed frontend/.env points Vite to the local Django API: VITE_API_BASE_URL=http://localhost:8000.
- Ran the pending Django migration assets.0006_asset_email so the Neon/PostgreSQL assets table matches the current Django model.
- Verified the local API after the fix: /health returned 200, /assets returned 200, and CORS allowed http://localhost:5173.

The Main Problem

Django was reachable, but the frontend could not use the data endpoint. A working /health endpoint only proves the server is alive. The frontend needs /assets, and that endpoint failed until the database schema was updated.

Before:

```
GET http://localhost:8000/health -> 200 OK
GET http://localhost:8000/assets -> 500 Internal Server Error
```

Database error:

```
column assets_asset.email does not exist
```

Fix applied:

```
cd backend
python manage.py migrate
```

Result:

```
GET http://localhost:8000/health -> 200 OK
GET http://localhost:8000/assets -> 200 OK
```

Local vs Production

Running Local

Frontend runs with Vite on localhost, usually http://localhost:5173. Backend runs with Django runserver on http://localhost:8000. The frontend uses VITE_API_BASE_URL=http://localhost:8000. Django must allow localhost and 127.0.0.1 in DJANGO_ALLOWED_HOSTS. CORS must allow the Vite origin.

Running on Production

Backend runs on the deployed domain, for example the DigitalOcean app host. Production must include that domain in DJANGO_ALLOWED_HOSTS. The deployed frontend must use the production backend URL in VITE_API_BASE_URL. Production environment values should be configured in the hosting platform, not committed to Git.

Important Environment Variables

- backend/.env: DJANGO_ALLOWED_HOSTS controls which hostnames Django will accept.

- backend/.env: BACKEND_CORS_ORIGINS controls which frontend URLs can call the backend from a browser.
- frontend/.env: VITE_API_BASE_URL controls where the React app sends API requests.
- backend/.env: DATABASE_URL controls which PostgreSQL/Neon database Django uses.

Local backend example:

```
DJANGO_ALLOWED_HOSTS=seashell-app-ksiia.ondigitalocean.app,localhost,127.0.0.1,[::1]  
BACKEND_CORS_ORIGINS=http://localhost:5173,http://127.0.0.1:5173
```

Local frontend example:

```
VITE_API_BASE_URL=http://localhost:8000
```

How To Run Locally

Use two terminals: one for Django and one for Vite.

Terminal 1 - Backend

```
cd /c/Users/ENVICOMM/Desktop/invent/backend  
source .venv/Scripts/activate  
python manage.py migrate  
python manage.py runserver 8000
```

Terminal 2 - Frontend

```
cd /c/Users/ENVICOMM/Desktop/invent/frontend  
npm run dev
```

How To Verify The Connection

- Open <http://localhost:8000/health>. It should show {"status": "ok"}.
- Open <http://localhost:8000/assets>. It should return JSON data, not a Django error page.
- Open the Vite URL, usually <http://localhost:5173>, then hard refresh with Ctrl + Shift + R.
- In the browser Network tab, confirm requests go to <http://localhost:8000/assets>, not an old /api/tickets/... path.

Plain-English Summary

The frontend and backend were not broken as separate apps. The local backend needed the right local host permissions, and the database needed the latest migration. Once the local host settings and database schema matched the code, the frontend could connect to Django correctly.